

Day 4: Hands-on Subnetting Practical (1 hour): Divide a network into subnets and assign IP addresses to devices. Task: Document the steps to create a subnet plan.

Task 1 :- Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

ANS :-

- **Subnet Mask for all 4: 255.255.255.192 (or /26)**

Subnet 1:

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63

Subnet 2:

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127

Subnet 3:

- Network Address: 192.168.10.128
- Broadcast Address: 192.168.10.191

Subnet 4:

- Network Address: 192.168.10.192
- Broadcast Address: 192.168.10.255

Task 2 :- For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

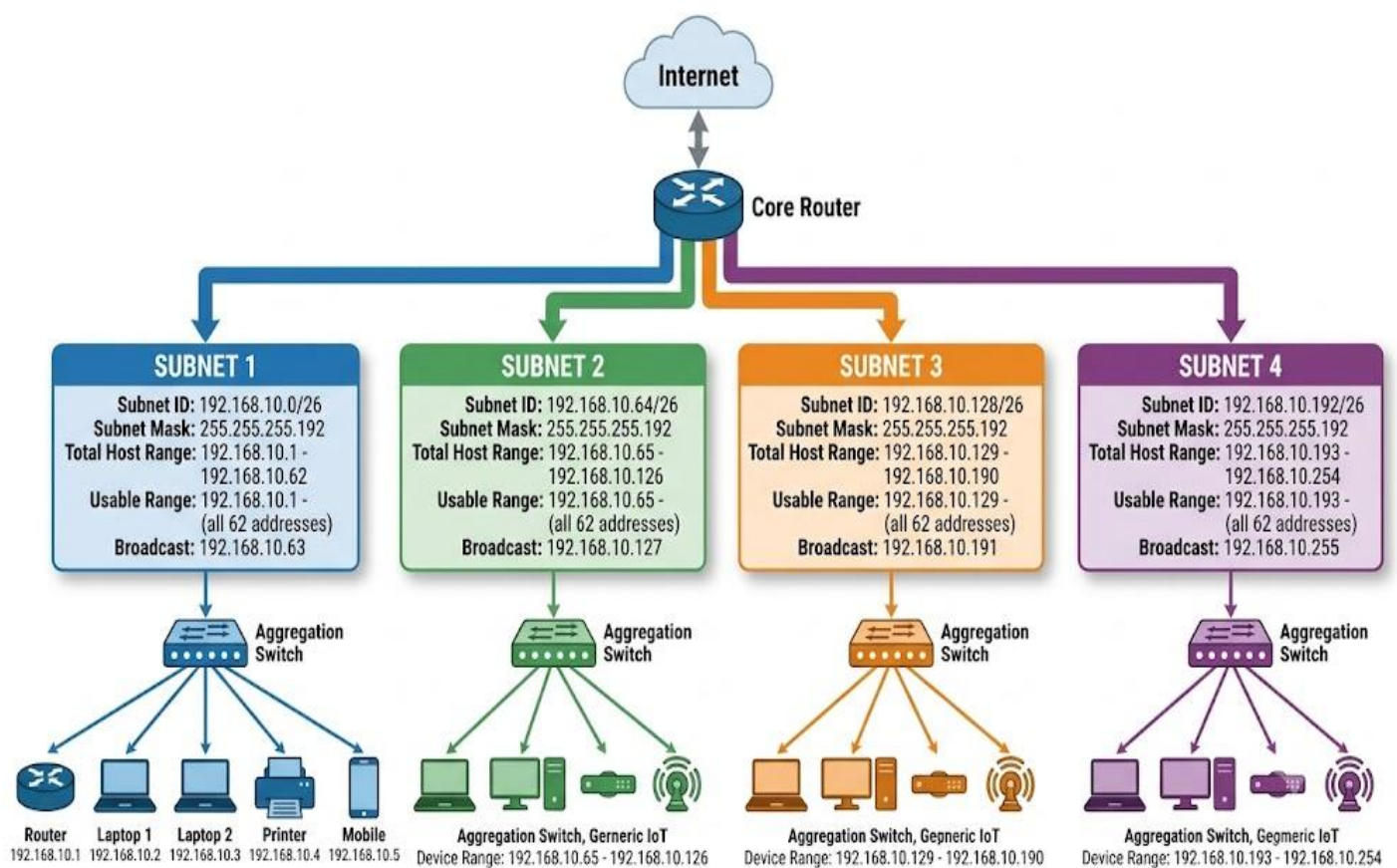
ANS :-

Using Subnet 1 (valid IPs: 192.168.10.1 to 192.168.10.62) :

- **Router** : 192.168.10.1
- **Laptop 1** : 192.168.10.2
- **Laptop 2** : 192.168.10.3
- **Printer** : 192.168.10.4
- **Mobile** : 192.168.10.5

Task 3 :- Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

ANS :-



Task 4 :- Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

ANS :-

1. Subnet Mask

- You need 4 subnets.
- $2 \times 2 = 4$, so borrow 2 bits.
- Add 2 to the original /24: $24 + 2 = /26$ (or 255.255.255.192).

2. Usable Hosts :-

- Each subnet gets 64 total IPs ($2^6 = 64$).
- Remove 1 for Network Name and 1 for Broadcast.
- $64 - 2 = 62$ usable devices per subnet.

3. Finding the IP Ranges

- Jump by 64s:
 - Subnet 1: 0 to 63
 - Subnet 2: 64 to 127
 - Subnet 3: 128 to 191
 - Subnet 4: 192 to 255

4. Assigning IPs (Subnet 1)

- First number (0) = Network ID.
- Last number (63) = Broadcast.
- Middle numbers (1 to 62) go to devices:
 - .1 Router
 - .2 Laptop 1
 - .3 Laptop 2
 - .4 Printer
 - .5 Mobile